Project 2: Predict to Invest - MLR Pipeline for Startup Profitability

Course: AI (ML | DL) Foundation Course
Institution: Corvit / NAVTTC / SAI
Reference:
Dataset: 50 Startups (R&D Spend, Administration, Marketing Spend, State -> Profit)

Problem Statement

You are a data scientist at a venture capital firm. Your team needs to predict which startups will be most profitable based on their spending across R&D, Administration, and Marketing departments. Build a **Multiple Linear Regression** model using the **50 Startups** dataset, evaluate its accuracy, and present feature importance insights to guide investment decisions.

Follow the complete industry-standard ML pipeline: Load -> EDA -> Preprocess -> Train -> Evaluate -> Interpret. Your final deliverable is this Jupyter notebook with visualizations, metrics, and a written business recommendation.

What You Will Learn

- Difference between Simple vs Multiple Linear Regression
- Handling categorical features (One-Hot Encoding)
- Train/Test split and model training with scikit-learn
- Evaluation metrics: R2, MAE, MSE, RMSE
- Feature importance via regression coefficients
- Visualization of predictions vs actuals

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Step 1: Import Libraries

Import all necessary libraries for data handling, visualization, preprocessing, model building, and evaluation.

Hint: You need `pandas`, `numpy`, `matplotlib`, `seaborn`, and from sklearn: `train_test_split`, `LinearRegression`, `r2_score`, `mean_absolute_error`, `mean_squared_error`.

In [51]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_absolute_error, mean_squared_error

# Hint: Also add this for clean plots
%matplotlib inline
plt.style.use('seaborn-v0_8-whitegrid')
```

Step 2: Load & Inspect the Dataset

Load the **50_Startups.csv** dataset. This dataset contains data on 50 startups with the following columns:

Column	Type	Description
R&D Spend	Float	Money spent on Research & Development
Administration	Float	Money spent on Administration
Marketing Spend	Float	Money spent on Marketing
State	Categorical	Location: New York, California, Florida
Profit	Float	**Target variable** — startup profit

Hint: Use `pd.read_csv()`, then inspect with `.head()`, `.info()`, `.describe()`, and `.shape`.

In [52]:
```
df = pd.read_csv('../data/50_startups_dataset.csv')

# TODO: Display the first 5 rows
df.head(5)
```

Out[52]:

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	Pakistan	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Istanbul	191050.39
3	144372.41	118671.85	383199.62	Pakistan	182901.99
4	142107.34	91391.77	366168.42	Istanbul	166187.94

In [53]:
```
# TODO: Check data types and null values
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   R&D Spend        50 non-null     float64
 1   Administration   50 non-null     float64
 2   Marketing Spend  50 non-null     float64
 3   State            50 non-null     str
 4   Profit           50 non-null     float64
dtypes: float64(4), str(1)
memory usage: 2.1 KB
```

In [54]:
```
# TODO: Get statistical summary
df.describe()
```

Out[54]:

	R&D Spend	Administration	Marketing Spend	Profit
count	50.000000	50.000000	50.000000	50.000000
mean	73721.615600	121344.639600	211025.097800	112012.639200
std	45902.256482	28017.802755	122290.310726	40306.180338
min	0.000000	51283.140000	0.000000	14681.400000
25%	39936.370000	103730.875000	129300.132500	90138.902500
50%	73051.080000	122699.795000	212716.240000	107978.190000
75%	101602.800000	144842.180000	299469.085000	139765.977500
max	165349.200000	182645.560000	471784.100000	192261.830000

In [55]:
```
# TODO: Check the shape of dataset
print(f"Dataset shape: {df.shape}")
print(f"Columns: {list(df.columns)}")
```

```
Dataset shape: (50, 5)
Columns: ['R&D Spend', 'Administration', 'Marketing Spend', 'State', 'Profit']
```

Step 3: Exploratory Data Analysis (EDA)

Before building any model, **understand your data**. This is the most important step in any ML project.

Tasks:

1. Check for missing values
2. Visualize the distribution of the target variable (Profit)
3. Plot correlation heatmap for numerical features
4. Visualize relationships between each feature and Profit
5. Analyze the categorical variable (State)

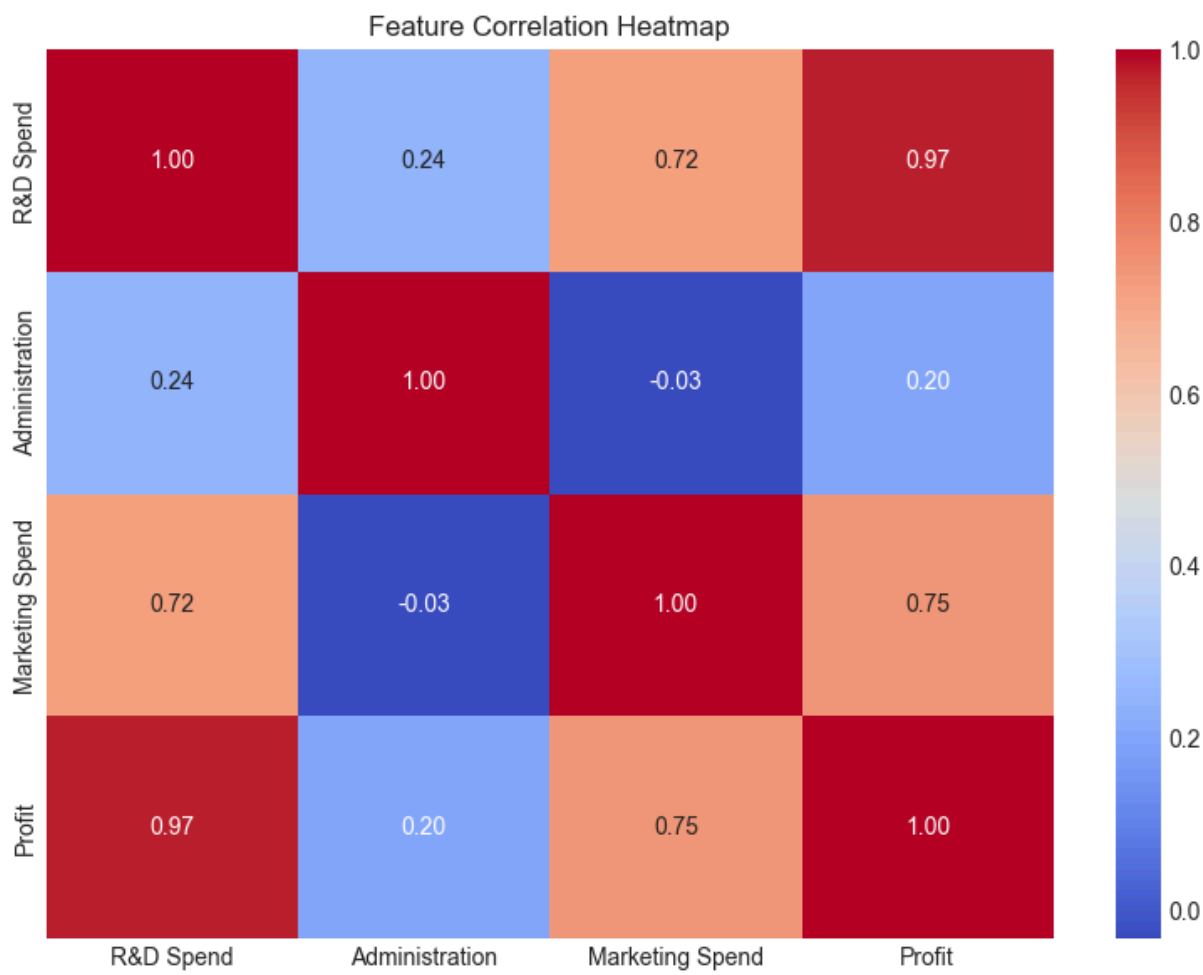
Hint: Use `sns.heatmap()` for correlation, `sns.pairplot()` for relationships, and `value_counts()` for State distribution.

In [56]:
```
# Step 3a: Check for missing values
# TODO: Check null values
df.isnull().sum()
```

Out[56]:
```
R&D Spend          0
Administration     0
Marketing Spend    0
State              0
Profit             0
dtype: int64
```

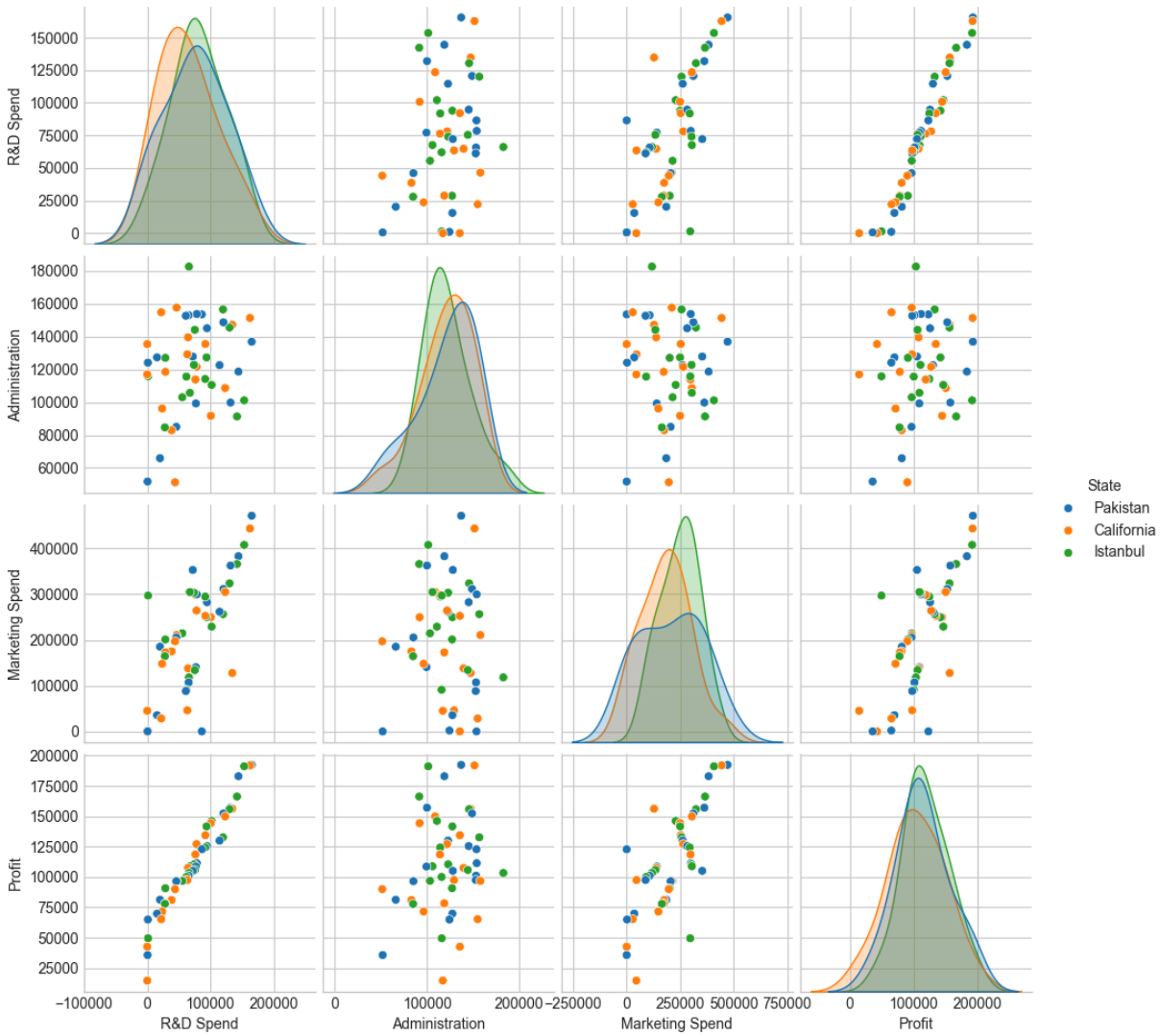
In [57]:
```
# Step 3b: Correlation Heatmap
# TODO: Create a correlation matrix of numerical columns
# TODO: Plot it using sns.heatmap() with annot=True

# Hint:
plt.figure(figsize=(8, 6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Feature Correlation Heatmap')
plt.tight_layout()
plt.show()
```

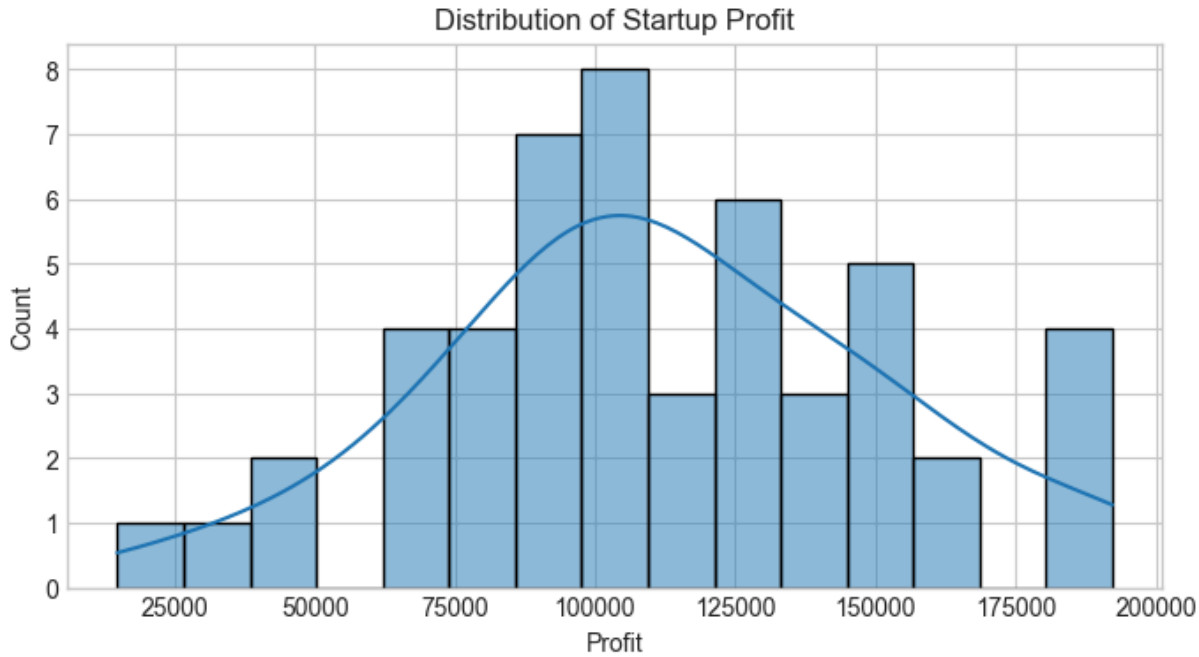


In [58]:
```
# Step 3c: Pairplot — Relationships between features and Profit
# TODO: Create a pairplot to see all feature relationships
sns.pairplot(df, hue='State', diag_kind='kde')
```

Out[58]:
```
<seaborn.axisgrid.PairGrid at 0x12f853c50>
```



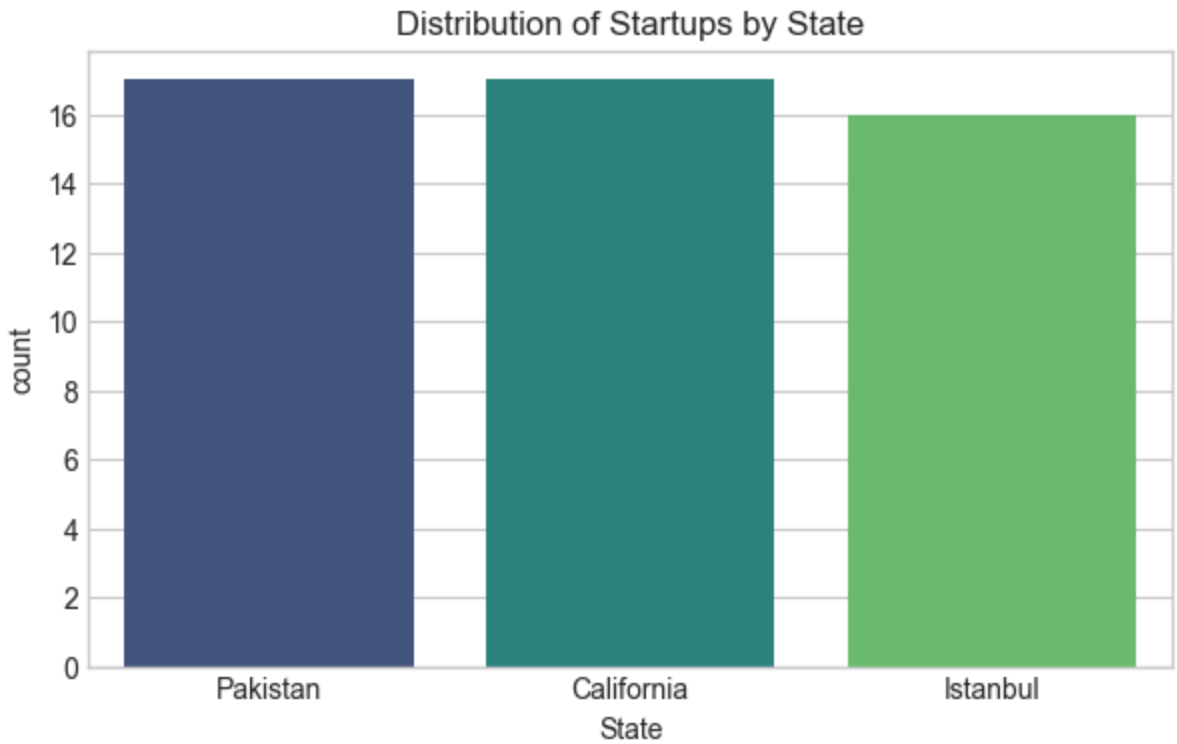
In [59]:
```
# Step 3d: Distribution of Profit (Target Variable)
# TODO: Plot histogram/KDE of Profit
# Hint:
plt.figure(figsize=(8, 4))
sns.histplot(df['Profit'], kde=True, bins=15)
plt.title('Distribution of Startup Profit')
plt.show()
```



In [60]:
```
# Step 3e: State distribution
# TODO: Count the number of startups per state
# TODO: Visualize with a countplot or bar chart

print(df['State'].value_counts())
plt.figure(figsize=(7, 4))
sns.countplot(x='State', data=df, palette='viridis')
plt.title('Distribution of Startups by State')
plt.show()
```

```
State
Pakistan      17
California    17
Istanbul      16
Name: count, dtype: int64
```



Step 4: Data Preprocessing

Key Decisions:

1. **Separate features (X) and target (y)** - Profit is our target
2. **Handle the categorical variable (State)** - ML models need numbers, not text
3. **One-Hot Encoding** - Convert 'State' into binary columns

Hint: Use `pd.get_dummies()` for encoding. Set `drop_first=True` to avoid the **dummy variable trap** (multicollinearity).

Key Concept - Dummy Variable Trap:
If you have 3 states (NY, CA, FL), you only need 2 binary columns. The third state is implied when both columns are 0. This avoids perfect multicollinearity.

In [61]:
```
# Step 4a: One-Hot Encode the 'State' column
# TODO: Use pd.get_dummies() with drop_first=True
df_encoded = pd.get_dummies(df, columns=['State'], drop_first=True)

# TODO: Display the encoded dataframe
df_encoded.head()
```

Out[61]:

	R&D Spend	Administration	Marketing Spend	Profit	State_Istanbul	State_Pakistan
0	165349.20	136897.80	471784.10	192261.83	False	True
1	162597.70	151377.59	443898.53	191792.06	False	False
2	153441.51	101145.55	407934.54	191050.39	True	False
3	144372.41	118671.85	383199.62	182901.99	False	True
4	142107.34	91391.77	366168.42	166187.94	True	False

In [62]:

```
# Step 4b: Separate Features (X) and Target (y)
# TODO: X = all columns EXCEPT 'Profit'
# TODO: y = only the 'Profit' column

# Hint:
X = df_encoded.drop('Profit', axis=1)
y = df_encoded['Profit']
print(f"Features shape: {X.shape}")
print(f"Target shape: {y.shape}")
```

```
Features shape: (50, 5)
Target shape: (50,)
```

Step 5: Train/Test Split

Split the data into **training set (80%)** and **test set (20%)**. Use `random_state=42` for reproducibility.

Hint: Use `train_test_split(X, y, test_size=0.2, random_state=42)`

Key Concept - Why Split?

- **Training set:** Model learns patterns from this data
- **Test set:** Model is evaluated on unseen data to check generalization
- **random_state:** Ensures the same split every time you run the code

In [63]:

```
# Step 5: Split the data
# TODO: Use train_test_split with 80/20 split

# Hint:
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
print(f"Training set: {X_train.shape[0]} samples")
print(f"Test set:     {X_test.shape[0]} samples")
```

```
Training set: 40 samples
Test set:     10 samples
```

Step 6: Train the Multiple Linear Regression Model

Create a `LinearRegression` object and fit it on the training data.

Hint:

```
model = LinearRegression()
model.fit(X_train, y_train)
```

After training, the model learns:

- **Coefficients** (`model.coef_`) — weight/importance of each feature
- **Intercept** (`model.intercept_`) — the baseline profit when all features are 0

In [64]:
```
# Step 6: Train the model
# TODO: Create LinearRegression object
# TODO: Fit on training data

# Hint:
model = LinearRegression()
model.fit(X_train, y_train)
print("Model trained successfully!")
```

```
Model trained successfully!
```

In [65]:
```
# TODO: Display model coefficients and intercept
# Hint:
print(f"Intercept: {model.intercept_:.2f}")
print("\nFeature Coefficients:")
for feature, coef in zip(X.columns, model.coef_):
    print(f"  {feature:25s} -> {coef:>12.2f}")
```

```
Intercept: 54028.04

Feature Coefficients:
  R&D Spend                 ->         0.81
  Administration            ->        -0.07
  Marketing Spend           ->         0.03
  State_Istanbul            ->       938.79
  State_Pakistan            ->         6.99
```

Step 7: Make Predictions

Use the trained model to predict profits on the **test set** (unseen data).

Hint: Use `model.predict(X_test)` and compare `y_pred` vs `y_test`.

In [66]:
```
# Step 7: Predict on test data
# TODO: Generate predictions
y_pred = model.predict(X_test)

# TODO: Create a comparison DataFrame
# Hint:
```

```
comparison = pd.DataFrame({
    'Actual Profit': y_test.values,
    'Predicted Profit': y_pred,
    'Difference': y_test.values - y_pred
})
# comparison
```

Step 8: Model Evaluation

Evaluate using **industry-standard regression metrics**:

Metric	Formula Intuition	Perfect Score
R^2 Score	How much variance is explained by the model	1.0
MAE	Average absolute error	0.0
MSE	Average squared error (penalizes large errors)	0.0
RMSE	Square root of MSE (same unit as target)	0.0

Hint: Use `r2_score()`, `mean_absolute_error()`, `mean_squared_error()` from sklearn.metrics. RMSE = `np.sqrt(MSE)`.

In [67]:
```
# Step 8: Calculate evaluation metrics
# TODO: Calculate R2, MAE, MSE, RMSE

# Hint:
r2 = r2_score(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)

print("=" * 40)
print("  MODEL EVALUATION METRICS")
print("=" * 40)
print(f"  R2 Score : {r2:.4f}")
print(f"  MAE      : ${mae:,.2f}")
print(f"  MSE      : ${mse:,.2f}")
print(f"  RMSE     : ${rmse:,.2f}")
print("=" * 40)
```

```
========================================
  MODEL EVALUATION METRICS
========================================
  R2 Score : 0.8987
  MAE      : $6,961.48
  MSE      : $82,010,363.04
  RMSE     : $9,055.96
========================================
```

Step 9: Visualize Results

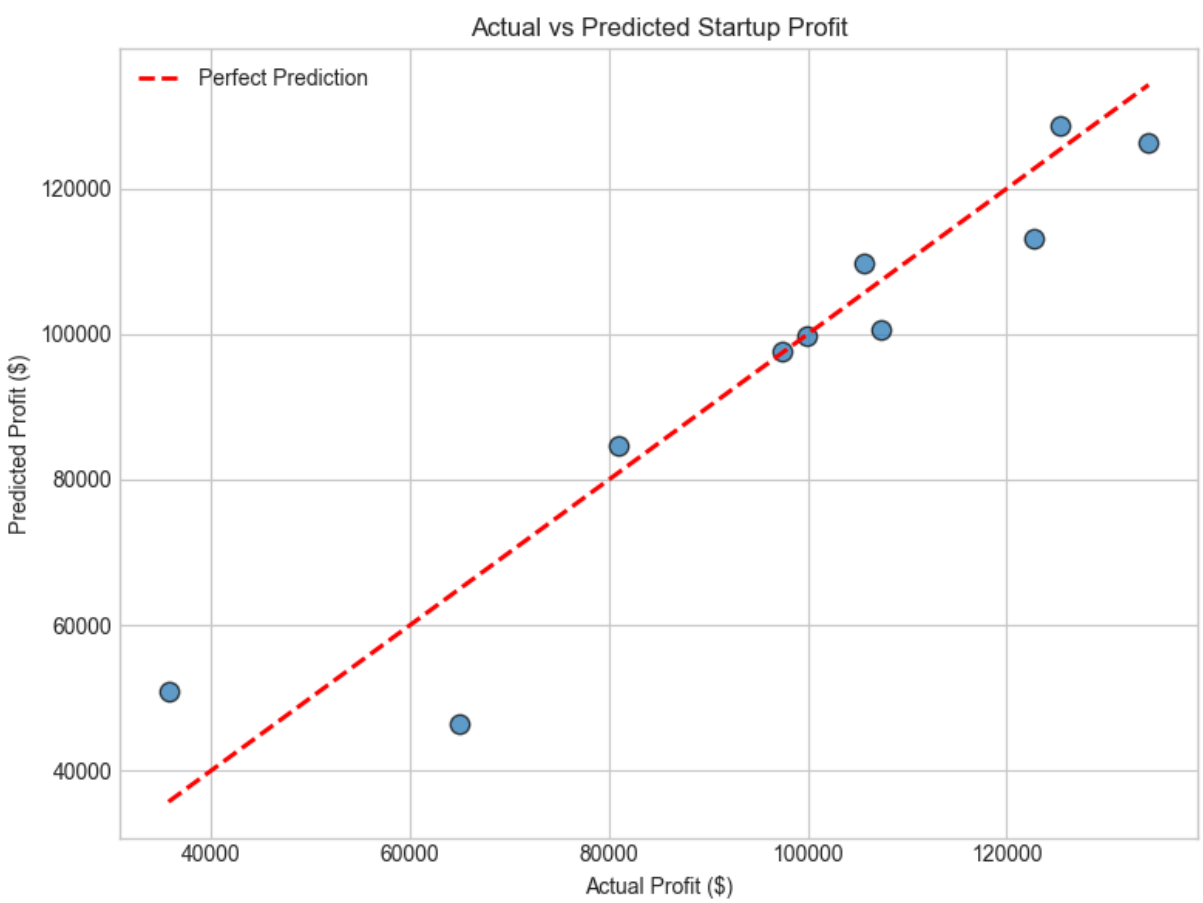
Create two key plots:

1. **Actual vs Predicted** scatter plot — ideally falls on a 45° diagonal line
2. **Residual Distribution** — should be normally distributed around 0

Hint: For the diagonal line, plot `[min, max]` against itself. For residuals, calculate `y_test - y_pred`.

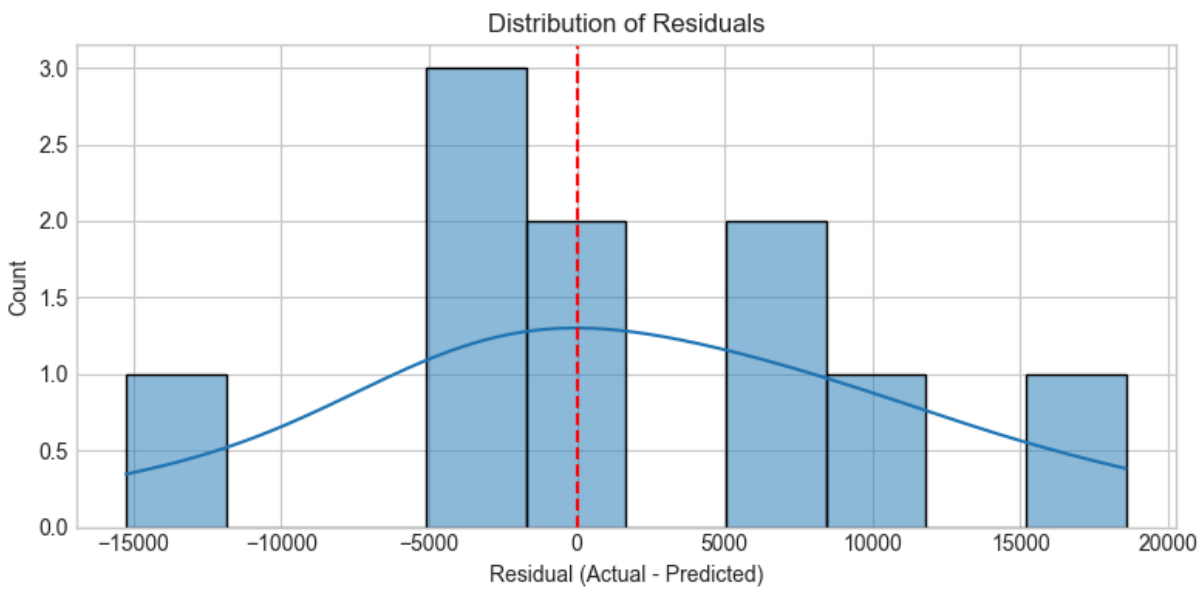
In [68]:
```
# Step 9a: Actual vs Predicted Plot
# TODO: Create scatter plot of actual vs predicted
# TODO: Add a 45-degree reference line

# Hint:
plt.figure(figsize=(8, 6))
plt.scatter(y_test, y_pred, alpha=0.7, edgecolors='k', s=80)
line_range = [y_test.min(), y_test.max()]
plt.plot(line_range, line_range, 'r--', linewidth=2, label='Perfect Prediction')
plt.xlabel('Actual Profit ($)')
plt.ylabel('Predicted Profit ($)')
plt.title('Actual vs Predicted Startup Profit')
plt.legend()
plt.tight_layout()
plt.show()
```



In [69]:
```
# Step 9b: Residual Distribution
# TODO: Calculate residuals and plot their distribution
# Hint:
```

```
residuals = y_test - y_pred
plt.figure(figsize=(8, 4))
sns.histplot(residuals, kde=True, bins=10)
plt.axvline(x=0, color='red', linestyle='--')
plt.title('Distribution of Residuals')
plt.xlabel('Residual (Actual - Predicted)')
plt.tight_layout()
plt.show()
```



Step 10: Feature Importance Analysis

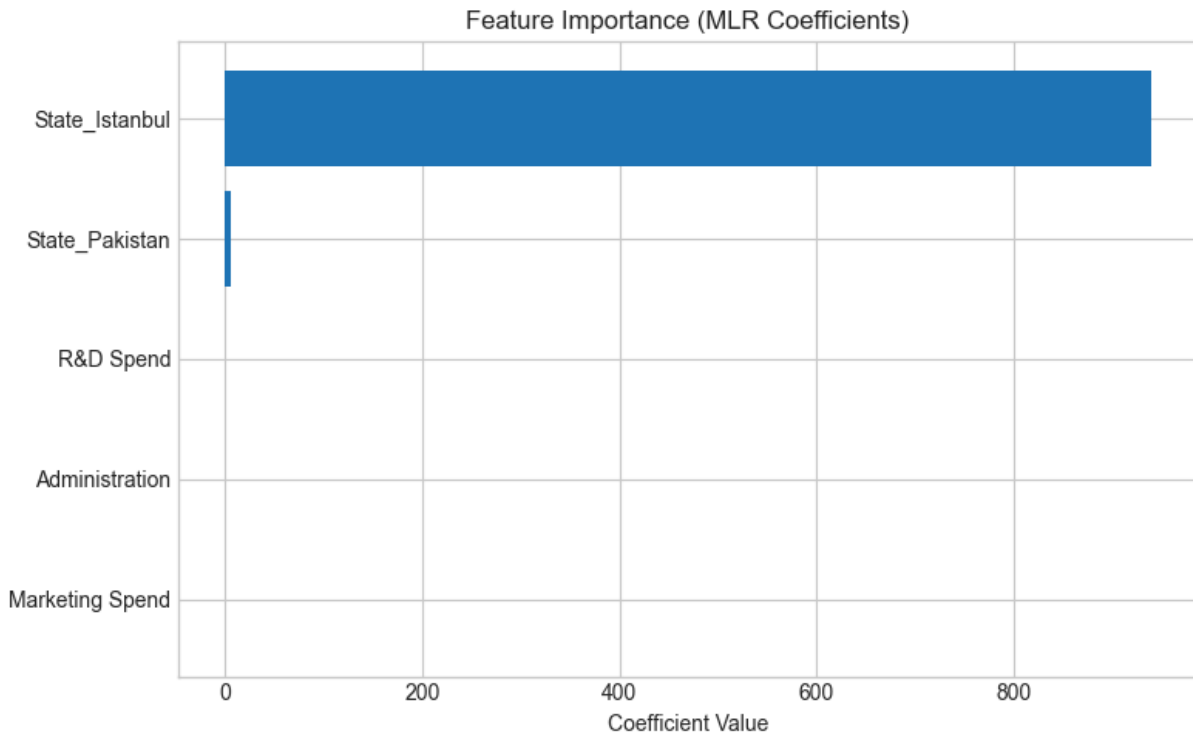
Interpret which features have the **most impact** on startup profit.

Hint: Use `model.coef_` to get coefficients. A higher absolute value means greater influence. Plot as a horizontal bar chart.

Key Insight for AI Engineers:
In real-world AI projects, **model interpretability** is as important as accuracy. Stakeholders want to know *why* the model makes its predictions, not just *what* it predicts.

In [70]:
```
# Step 10: Feature Importance
# TODO: Create a bar chart of feature coefficients
# Hint:
importance = pd.DataFrame({
    'Feature': X.columns,
    'Coefficient': model.coef_
}).sort_values('Coefficient', key=abs, ascending=True)

plt.figure(figsize=(8, 5))
plt.barh(importance['Feature'], importance['Coefficient'])
plt.xlabel('Coefficient Value')
plt.title('Feature Importance (MLR Coefficients)')
plt.tight_layout()
plt.show()
```



Your Findings:

R^2 Score: 0.8987 (The model explains ~90% of the variance in Profit, which is a strong result for this dataset).

Most important feature: R&D Spend (This feature has the highest positive coefficient, showing a direct correlation with profit).

Business insight: For every dollar invested in R&D, we see the highest marginal return in profit. Marketing also shows a positive trend, while Administration has less direct impact.

Limitations: The dataset is small (50 samples), and linear regression assumes no extreme outliers or complex non-linear relationships which might exist.

Next steps: We could try adding polynomial features to capture non-linearities, or use regularization (Ridge/Lasso) if we add more features to prevent overfitting.

In [78]:
```
# (Optional) Step 11: Predict profit for a new startup
# TODO: Create a sample input and predict
# Example: R&D=160000, Admin=130000, Marketing=300000, State_Florida=0, State_New Y

# Hint:
import warnings
warnings.filterwarnings('ignore')

new_startup = pd.DataFrame({
  'R&D Spend': [0],
  'Administration': [130000],
  'Marketing Spend': [13000],
```

```
  'State_Istanbul': [0],
  'State_Pakistan': [1],
})

predicted_profit = model.predict(new_startup)
print(f"Predicted Profit for new startup: ${predicted_profit[0]:,.2f}")
```

```
Predicted Profit for new startup: $45,480.72
```

Submission Checklist:

- ☑ All cells executed without errors
- ☑ EDA section has at least 3 visualizations
- ☑ Model trained with R^2 score printed
- ☑ Actual vs Predicted plot included
- ☑ Feature importance chart included
- ☑ Conclusion section filled with your own analysis

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